Explanation of the Research Component of the Academic Master's Program in Mechanical Engineering

and Its Substantial Comparability to the UQ Master of Philosophy (MPhil)

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Institution	China University of Mining and Technology	**Program**	Full-time Academic Master's in Mechanical Engineering (2022 cohort)

Program structure and research character. CUMT's official 2022 training scheme identifies Mechanical Engineering as a three-year academic master's program. Coursework is normally completed within the first academic year and requires at least 22 credits. Mr Hu's official English academic record is expressly titled *"Course Learning Period"* and records 23 accumulated credits, which explains why the thesis assessment is not shown on that course transcript.

- After the limited coursework stage, the remainder of the program (approximately two years) is devoted primarily to supervised research, thesis development, academic publications and other research outputs.
- Compulsory research milestones include a literature review and thesis proposal, at least 8 credits in research competence and innovation (including academic activities, research practice, independent research, papers and patents), a mid-term review/pre-defence, thesis examination and an oral defence.
- Mr Hu completed a substantial thesis of approximately 150 pages. His thesis was rated Outstanding by CUMT, evidencing achievement above the normal thesis requirement;

Comparison with the UQ MPhil. UQ describes its MPhil (program 7500) as a 1.5- to 2-year full-time graduate research degree involving a significant independent research project, a 40,000-word thesis and an oral examination. Although the degree title and transcript conventions differ, the CUMT program contains the same core features of a research master's degree: formal supervision, a substantial independent research period, assessed research milestones, a major thesis, and an oral defence. It is therefore substantially comparable to the UQ MPhil in research character and preparation for doctoral study.

Official sources:

1. CUMT Graduate School webpage for the 2022 training schemes: https://gs.cumt.edu.cn/info/1034/3249.htm
2. Official CUMT 2022 Academic Master's Training Scheme (Mechanical Engineering, printed pp. 73-77, please open with **Chrome PDF viewer**): https://gs.cumt.edu.cn/_local/9/59/D4/1430A0EB4506A480FBD2814A439_17C43BF5_383195.pdf?e=.pdf
3. UQ Master of Philosophy (Program 7500): https://study.uq.edu.au/study-options/programs/master-philosophy-7500

Certification by China University of Mining and Technology

Name and title of authorised officer:

Signature: Yanfei Deng

Date: 2026.10.10

中国矿业大学 研究生院

China University of Mining and Technology

Training Scheme for Academic Master's Degree Students (2022 Edition)

English Translation of the Mechanical Engineering Section
First-Level Discipline Code 0802 | Source printed pages 73-77

Original document	CUMT Training Scheme for Academic Master's Degree Students (2022 Edition)
Official webpage	https://gs.cumt.edu.cn/info/1034/3249.htm
Official source PDF (please open with Chrome PDF viewer)	https://gs.cumt.edu.cn/__local/9/59/D4/1430A0EB4506A480FBD2814A439_17C43BF5_383195.pdf?e=.pdf
Translated section	0802 Mechanical Engineering - School of Mechatronic Engineering, pages 73-77

Translation Certification (to be completed by CUMT or an authorised translator)

I certify that the following is a true and accurate English translation of the specified pages of the official Chinese-language document identified above.

Signature	Yanfei Deng
Date	2026.10.10
Official seal / stamp	

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中国矿业大学 研究生院

0802 MECHANICAL ENGINEERING (FIRST-LEVEL DISCIPLINE)

Scope of Application	
005 School of Mechatronic Engineering	080200 Mechanical Engineering

I. Introduction to the Discipline

The Mechanical Engineering discipline at China University of Mining and Technology (CUMT) was established in 1950, originally as Mining Machinery Engineering. It was authorised in 1981 as one of the first doctoral degree-granting disciplines. In 1997, the discipline was adjusted to Mechanical Design and Theory; in 1999, it was approved to establish a postdoctoral research station in Mechanical Engineering; in 2006, it was approved as a first-level discipline doctoral degree program; in 2007, Mechanical Design and Theory was designated a National Key Discipline; in 2008, Mechanical Engineering was designated a Jiangsu Provincial Key First-Level Discipline and included in the Ministry of Education's "985 Project" Innovation Platform for Advantageous Disciplines; in 2009, it was designated a Jiangsu Provincial cultivation site for a National Key First-Level Discipline; and since 2011, it has been included for three consecutive phases in the Jiangsu Higher Education Advantageous Discipline Construction Program. The discipline currently supports the Provincial-Ministerial Collaborative Innovation Center for Intelligent Mining and Excavation Equipment, a "111 Project" talent-introduction base, the Jiangsu Engineering Technology Research Center for Intelligent Fully Mechanized Mining and Roadheading Equipment, and the Jiangsu Key Laboratory of Mine Electromechanical Equipment.

The discipline aims to build a first-class Mechanical Engineering discipline distinguished by mining machinery and construction machinery. Through many years of development, it has established distinctive strengths in intelligent mining and excavation equipment, high-efficiency mine hoisting and transportation, transmission and control of heavy machinery, and mining robotics.

The discipline has a strong faculty, including 32 professors, 1 research professor and 46 associate professors. Among them are 35 doctoral supervisors and 52 master's supervisors. Faculty members include one recipient of the Ho Leung Ho Lee Foundation Prize, one recipient of the China Youth Science and Technology Award, one selected for the New Century National Hundred-Thousand-Ten Thousand Talents Project, three recipients of the State Council Special Government Allowance, one selected for the China Association for Science and Technology Young Talent Support Project, four selected for the Ministry of Education's New Century Excellent Talents Program, one recipient of the Jiangsu Provincial Outstanding Youth Science Foundation, three recipients of the Sun Yueqi Youth Science and Technology Award, three recipients of the National Coal Youth Science and Technology Award, nineteen selected for the Jiangsu "333" Talents Program, and forty-seven selections in other provincial and ministerial talent programs. The discipline has one Ministry of Education Innovation Team, two Jiangsu University Excellent Innovation Teams, one Jiangsu "Qinglan Project" Teaching Team, and one Jiangsu "Six Talent Peaks" Innovation Team.

During China's 13th Five-Year Plan period, the discipline focused on the national strategic need for deep-earth resource development. It undertook six National Key Research and Development Program projects, more than sixty National Natural Science Foundation of China projects, more than thirty provincial and ministerial research projects, and more than 350 enterprise-commissioned projects, with cumulative research funding received of nearly RMB 150 million. Major breakthroughs were achieved in large mine-hoisting equipment, long-distance high-capacity belt conveying equipment, intelligent coal-mining equipment, and condition-monitoring technology for mine equipment. These achievements generated independent intellectual property, including more than 200 granted overseas invention patents and more than 500 granted Chinese invention patents. The discipline received one Second Prize of the National Technological Invention Award, one Second Prize of the National Science and Technology Progress Award, two First Prizes and two Second Prizes of the Jiangsu Science and Technology Award, and two Second Prizes of the Ministry of Education Science and Technology Progress Award. Important fundamental research results were published in leading journals including Chinese Journal of Mechanical Engineering, IEEE/ASME Transactions on Mechatronics, Mechanical Systems and Signal Processing, and Nonlinear Dynamics.

During the 13th Five-Year Plan period, the discipline enrolled a total of 173 doctoral students, of whom 93 were awarded the Doctor of Engineering degree.

II. Major Research Areas

The discipline conducts fundamental and key-technology research in Mechanical Engineering. Its major research areas include:

1. Intelligent mining equipment
2. Mechanical transmission and control
3. Robotics
4. Modern mechanical design theory and methods
5. Tribology
6. Mechanical system dynamics
7. Intelligent health management of machinery
8. Mechatronics / mechanical electronics
9. Nontraditional machining
10. Intelligent manufacturing
11. Special-purpose vehicles

III. Training Objectives

The program aims to develop high-level research and management professionals with a solid foundation, comprehensive qualities, strong academic research capability and a certain level of innovation capability. The basic requirements are as follows:

1. Students shall master the basic theories of Marxism, uphold the leadership of the Communist Party of China, love the motherland, abide by laws and regulations, demonstrate integrity, fairness, scientific rigour and proper academic conduct, and possess a strong sense of social responsibility, sound academic ethics, and an innovative and entrepreneurial spirit.
2. Students shall master a solid theoretical foundation and systematic specialised knowledge in Mechanical Engineering; understand the current state and development directions of their research field; possess sound innovative awareness and the ability to conduct scientific research; and be capable of independently undertaking specialised technical and management work in Mechanical Engineering.
3. Students shall master a foreign language, be able to read professional literature in that language with relative proficiency, possess an international outlook, and have a certain ability in foreign-language writing and international academic communication.
4. Students shall respect science and possess a spirit of exploration and dedication to scientific research, a rigorous research style, and sound teamwork capability.
5. Students shall have sound physical and mental health.

IV. Standard Duration and Maximum Period of Study

The standard duration of study is three years, and the maximum period of study is four years (including approved suspension of study). A student who has held master's student status for at least two years may apply for early graduation in accordance with the CUMT Regulations on Early Graduation for Graduate Students.

V. Curriculum

Graduate courses are divided into general compulsory courses, discipline compulsory courses and elective courses. Course learning should normally be completed within the first academic year, and the course-learning component must total no fewer than 22 credits.

Students admitted from a different first-level discipline shall, as required by the supervisor, complete two core undergraduate courses in the discipline (or major) and obtain passing or higher grades. Students may select additional courses according to their knowledge structure and research needs. Self-selected and remedial courses are graded but do not carry credits.

Under the guidance of the supervisor, master's students may take graduate courses offered by other disciplines. Students are encouraged to take integrated master's-doctoral courses or other doctoral-level courses in the discipline; after passing these courses, they may apply for exemption from the corresponding courses during doctoral study.

Course Structure for the Academic Master's Degree

Category	Course code	Course title	Hours / Credits / Semester
General compulsory	140001C6	Introduction to Dialectics of Nature	18 / 1 / 2
General compulsory	140002C6	Studies on the Theory and Practice of Socialism with Chinese Characteristics for a New Era	36 / 2 / 2
General compulsory	100001C6	Basic Academic English Communication [Option Group A]	32 / 2 / 1
General compulsory	100002C6	Selected Readings in Scientific and Technical Literature [Option Group A]	32 / 2 / 1
General compulsory	100003C6	Master's English: Academic Paper Writing [Option Group B]	32 / 2 / 2
General compulsory	100004C6	Master's English: Academic Translation Practice [Option Group B]	32 / 2 / 2
Discipline compulsory	050001S6	Frontiers in Mechanical Engineering (Integrated Master's-Doctoral Course)	32 / 2 / 1
Discipline compulsory	050002S6	Academic Paper Writing Guidance in Mechanical Engineering (Integrated Master's-Doctoral Course)	16 / 1 / 1
Discipline compulsory	080014B6	Advanced Engineering Mathematics	32 / 2 / 1
Discipline compulsory	050003S6	Modern Testing Systems and Technology (English-taught)	32 / 2 / 1
Discipline compulsory	050004S6	Mechanical Dynamics	32 / 2 / 1
Discipline elective	050005S6	Mining Equipment and Intelligent Technologies (Integrated Master's-Doctoral Course)	48 / 3 / 2
Discipline elective	050006S6	Advanced Robotics (English-taught; Integrated Master's-Doctoral Course)	32 / 2 / 2
Discipline elective	050007S6	Frontiers in Tribology (Integrated Master's-Doctoral Course)	32 / 2 / 2
Discipline elective	050008S6	Intelligent Condition Monitoring and Fault Diagnosis Methods (Integrated Master's-Doctoral Course)	32 / 2 / 2
Discipline elective	050009S6	Modeling, Control and Intelligent Optimization of Manufacturing Processes (Integrated Master's-Doctoral Course)	32 / 2 / 2
Discipline elective	050010S6	Complex Robot Modeling and Intelligent Control	32 / 2 / 2
Discipline elective	050011S6	Nontraditional Machining Technology	32 / 2 / 2
Discipline elective	050012S6	Intelligent Sensing and Machine Vision	32 / 2 / 2
Discipline elective	050013S6	Microelectromechanical Systems Technology (Bilingual)	32 / 2 / 2
Discipline elective	050014S6	Analysis and Design of Modern Manufacturing Systems (English-taught)	32 / 2 / 2
Discipline elective	050015S6	Principles and Applications of Virtual Reality	32 / 2 / 2

Option Group A: choose one of 100001C6 and 100002C6. Option Group B: choose one of 100003C6 and 100004C6.
Note: At least one integrated master's-doctoral course must be selected.

VI. Other Requirements

1. New Student Orientation

This component is completed during the first week after enrolment and carries 1 credit. It combines patriotic education with education on identification with and commitment to the University, helping new students establish firm ideals and convictions, strengthen their commitment to the country and the University, adapt quickly to graduate research and academic life, and complete the transition to their new role. The main content includes education on University history and traditions; ethics and core socialist values; interpretation of graduate training policies and routine administration; scientific ethics and academic conduct; discipline-specific study; mental health; and safety education.

2. Individual Training Plan

Within one month after enrolment, each master's student shall formulate an individual course and training plan under the guidance of the supervisor and with participation from the supervisory group. The plan shall be approved by the responsible leader of the School and filed with the School before the tenth week of the first semester for monitoring and assessment. If changes are required during implementation owing to special circumstances, the revised plan shall be approved and signed by the supervisor and re-filed with the School.

3. Literature Review and Thesis Proposal

Within the standard duration of study, each master's student has two opportunities to complete the thesis proposal: the first at the end of the third semester and the second in the middle of the fourth semester. Both are conducted in a centralised format. Only students who pass the proposal may continue their master's thesis research. A student who fails both proposal attempts or does not undertake the proposal will have graduation automatically postponed and may, upon individual application, undertake one additional proposal in the fifth semester. A student who fails or does not undertake this additional proposal will be subject to automatic program diversion/exit. Detailed requirements are governed by the CUMT Regulations on Graduate Thesis Topic Selection.

Passing both the literature review and thesis proposal components carries 1 credit.

4. Research Competence and Innovation Capacity

The Research Competence and Innovation Capacity component requires a total of at least 8 credits:

(1) Research competence: at least 4 credits, assessed primarily through academic activities and thematic seminars. The relevant requirements are implemented in accordance with the CUMT Regulations on the Basic Requirements and Assessment of Research Competence for Full-time Academic Graduate Students.

(2) Innovation capacity: at least 2 credits, assessed primarily through academic papers, research practice, independent research, patent inventions, and extracurricular innovation competitions. The relevant requirements are implemented in accordance with the CUMT Regulations on the Basic Requirements and Assessment of Innovation Capacity for Full-time Academic Graduate Students.

5. Academic Norms and Research Ethics

The assessment of academic norms and research ethics is conducted after completion of the degree thesis and before submission for thesis examination. A 5,000-Chinese-character reflection on academic norms and research ethics shall be prepared with reference to the student's published papers, submitted papers, other research outputs, scientific ethics and academic conduct, and shall be reviewed by the supervisor. After approval, 1 credit is awarded and the thesis may be submitted for examination. A thesis may not be submitted if this assessment has not been completed or has not been approved.

6. Mid-term Review (Pre-defence)

At the mid-point of study, the student is comprehensively assessed with respect to completion of coursework, the literature review and proposal, progress on the degree thesis, work attitude and level of effort. Within the standard duration, each master's student has two opportunities for the mid-term review: the first at the end of the fifth semester and the second at the beginning of the sixth semester, both conducted in a centralised format. The thesis may be submitted for examination only after the mid-term review has been passed. A student who fails both reviews or does not attend will have graduation automatically postponed and may, upon individual application, attend one additional review in the seventh semester. A student who fails or does not attend this additional review will be subject to automatic program diversion/exit.